

# BHANU NAGPURE

Machine Learning & AI Engineer

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bhanu-dev82.github.io | Nagpur, India | Open to on-site / remote / hybrid

## SUMMARY

Machine-learning engineer with **international research experience** (Mitacs Globalink, Canada) and hands-on production AI: **95%-accuracy** text classification with BERT/Transformers, **on-device LLM inference** (LiteRT-LM, Gemma/Phi), and computer vision (MediaPipe, ML Kit, YOLOv5). Published in **IGI Global** and **ACM**; **2 patent applications (1 granted)**. Strong in Python, PyTorch/TensorFlow, and shipping AI into production mobile products.

## TECHNICAL SKILLS

**ML & Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers, BERT, Scikit-learn, fine-tuning, evaluation (accuracy/precision/recall)

**NLP & Generative AI:** Text classification, tokenization, Gemini API (Flash Lite), context-aware prompt/system design

**Computer Vision:** OpenCV, YOLOv5, MediaPipe FaceMesh, Google ML Kit, optimised inference pipelines

**On-Device AI:** Dual runtime (LiteRT-LM + GGUF), Gemma 4, Qwen3.5, Phi 3.5 Mini, SmoLLM, offline inference

**Languages:** Python (research), Java, Kotlin, Dart, C++, SQL

**MLOps & Deployment:** Flask/REST service deployment, Firebase, Git, CI/CD, model lifecycle

## PROFESSIONAL EXPERIENCE

### Devanshu Studios - Founder & AI Engineer

Aug 2025 - Present

*Independent · Remote*

- Built **Chakuli**, an on-device LLM chatbot running **Gemma 4, Qwen3.5, Phi 3.5 Mini and SmoLLM** via a dual **LiteRT + GGUF** runtime - full AI inference offline, zero cloud calls, preserving user privacy.
- Designed a **mastery-based adaptive practice engine** that detects weak topics and generates focused AI sessions; integrated a **Gemini AI tutor** with per-track system prompts.

### Athabasca University - ML Research Intern (Mitacs Globalink)

Aug 2024 - Oct 2024

*Edmonton, AB, Canada*

- Built a text-ideology classifier using **BERT/Transformers** in Java/Python reaching **95% prediction accuracy**; deployed as a **REST-backed web service** for researcher validation.

### BETIC-GHRCE - ML & Software Developer Intern

Oct 2022 - May 2024

*Nagpur, India*

- Applied **MediaPipe FaceMesh + Google ML Kit** to a medical diagnostic app, improving accuracy by **60%** and inference speed by **30%** through camera-pipeline optimisation.
- Contributed to a **granted patent** (TMJ Screening Tool, Pub. No. 202321022476) using the same on-device computer-vision stack.

### IIT Bombay - Design for Health Fellow

Mar 2022 - Oct 2022

*Mumbai, India*

- Led a team prototyping a xerostomia diagnostic device, reducing assessment time by **80%**; software layer on IoT hardware, presented at IIT Bombay.

## KEY PROJECTS

### Rote Playoffs - Agent Procedure Benchmark | Python, controlled A/B benchmarking, deterministic grading, telemetry

Sep 2026

- Measured procedure reuse in agentic LLM workflows across 6 tasks: **9.9x faster, 15.3x fewer output tokens**, 9.8/10 with-Play correctness; frozen ground truth, deterministic grader.

### Chakuli - On-Device AI Chatbot | LiteRT + GGUF, Gemma 4/Qwen3.5/Phi/SmoLLM, Kotlin, Compose

2025 - Present

- Offline AI assistant with zero-cloud LLM inference and an Agent Skills system; foreground WorkManager manages multi-GB model downloads with exponential backoff.

### Text Ideology Classifier | Python, Transformers/BERT, Java, Flask

2024

- 95% prediction accuracy**; deployed as a REST web service for researcher validation at Athabasca University.

### Autonomous Garbage Collection Robot | Python, YOLOv5, OpenCV, IoT

2023

- YOLOv5 computer-vision pipeline for autonomous waste detection - **60%** reduction in manual effort. [Published at ACM ICACS 2024, Hong Kong](#).

## EDUCATION

### G H Raisoni College of Engineering

Dec 2021 - Jun 2025

*B.Tech in Computer Science & Engineering | CGPA: 8.7 / 10.0*

*Nagpur, India*

Operating Systems, Computer Networks, DBMS, OOP, Data Structures & Algorithms

## PUBLICATIONS, PATENTS & AWARDS

- Journal:** "V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT." *IGI Global*, Mar 2024.
- Conference:** "Autonomous Floating Garbage Collection Device using Computer Vision and Robotics." *8th ICACS 2024, ACM, Hong Kong*, Oct 2024.
- Conference:** "Development of a Mobile Field Trip Application." *3rd Annual TRESL Lab International Symposium*, Athabasca, Canada, Jul 2024.
- Patent (Granted):** "Smart Chairside TMJ Examination & Muscle Activity Detector Tool," India (Pub. No. 202321022476).; **Patent (Pending):** "AI-Based Portable Device for Dental Disease Detection," DPIIT, India (App. No. 202421006731, Published Aug 2025).
- Awards:** 3rd Place · Avishkar 2024 | Runner-up · DIPEX 2023 & SHRISTI 2023 | BIREC E-YUVA Grant · Rs. 10 lakh seed funding | Winner · MEDHA 2022 & MEDIC-2022